

# Branch

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## 1. Feature Overview

Branch master (codes, name, address, IFSC-like metadata). **Not implemented** as a module. Accounts store free-text `branchCode` (often hardcoded `0001`).

**Status:** Stub / Planned

## 2. Business Purpose

Multi-branch operations, reporting by branch, validated branch codes in account numbers.

## 3. User Workflow

Planned: ADMIN maintains branches; account open selects branch from list.

## 4. Execution Flow

Today:

```
OpenAccountRequest.branchCode (string)
    ↓
AccountNumberGenerator.generate(branchCode, ...)
```

## 5. Database Tables

None. Planned: `branch`

## 6. REST APIs

None. Planned: `/branches` CRUD

## 7–13. Controllers / Services / Repositories / DTOs / Entities / Utils / Config

None

## 14. Validation Rules

Today: none beyond non-null string on open. Planned: FK exists and active.

## 15. Security Rules

Planned: ADMIN write; broader read

## 16–18. Exception / Logging / Audit

N/A

## 19. Testing Strategy

When built: reject unknown branchCode on open account

## 20. Future Extension Guide

Add `branch` entity; migrate existing `account.branch_code` values; wire Open Account / Customer create dialogs.

# Future Modification Guide

Requirement: Add branch master and validate on open

| Item    | Detail                                                                                                                               |
|---------|--------------------------------------------------------------------------------------------------------------------------------------|
| Files   | New <code>com/bankone/branch/**</code> ; <code>AccountServiceImpl.openAccount</code> ; Angular <code>account/customer</code> dialogs |
| Classes | <code>BranchController</code> , <code>BranchService</code> , <code>AccountServiceImpl</code>                                         |
| Methods | <code>openAccount()</code> — <code>branchRepository.findByCode</code> before generate                                                |
| Impact  | Account number generation; reports; employee home-branch later                                                                       |

### Call hierarchy (target)

```
AccountController.openAccount()  
  ↓  
AccountServiceImpl.openAccount()  
  ↓  
BranchService.requireActive(branchCode)  
  ↓  
AccountNumberGenerator.generate()  
  ↓  
AccountRepository.save()
```